08-11 第2篇

阅读理解（说明文） How Do Ants Find Their Way Home?

① You have probably seen it many times: a single black ant walking across the kitchen floor, carrying a piece of rice or a crumb of bread three times bigger than itself. You leave the room, come back ten minutes later, and suddenly there is **a long line of ants**, all walking carefully along the same path. By evening, the crumb and all its friends have disappeared. This small everyday scene hides a surprising question: **how does an ant, with a brain smaller than a grain of salt, find its way home and back again so accurately?**

② The short answer is: ants do not use one big skill — they use **several small skills, carefully combined**. Scientists have studied ants for more than a hundred years, and they now agree on three main tools that most ants use to navigate: **chemical signals**, **the sun**, and **simple memory of the landscape**.

③ The first tool is **chemical signals**, or "pheromones". When an ant walks, it can press its belly down and leave a thin invisible line of scent on the ground. This scent is very weak to us, but other ants can clearly smell it with their long bendy feelers, called antennae. When an ant finds food — a piece of biscuit, for example — it hurries back to the nest, leaving a pheromone trail behind. Back home, it touches its friends with its antennae to say, "I found something good." The friends follow the scent, and each of them leaves more pheromone on the way. In a short time, a thin line becomes a strong, confident line — which is why we see that neat "ant highway" across the floor.

④ Pheromones also fade slowly. If food runs out, fewer ants come back, fewer new trails are laid, and the old scent slowly disappears in a few hours. In this way, the ant colony quickly "forgets" an empty food spot and focuses on new ones. No ant is giving orders. The trail itself is the message.

⑤ The second tool is **the sun**. Many ants — especially desert ants — cannot always use pheromones, because hot sand makes the scent fade in a few seconds. Instead, they use the position of the sun in the sky. An ant's eyes can also see **polarised light** — something humans cannot see without special glasses — which allows the ant to sense the direction of the sun even when it is hidden by clouds. The ant's brain quietly counts **how many steps** it walks, and in which direction, so that when it needs to go home, it can walk almost in a straight line back to the nest. This behaviour is called "**path integration**". Clever experiments show that if scientists pick up an ant and move it to a new spot, the ant will still walk in the same direction and for the same distance it planned before — then stand there, confused, because home is no longer there.

⑥ The third tool is **landscape memory**. Many ants living near trees and gardens use small features of the land — a rock, a blade of grass, the shadow of a bush — as familiar "signposts". When they walk out to find food, they quickly learn these signposts, and when they turn back, they check each one in order, almost like a person reading the same street signs on the way home. Some researchers think an ant's brain keeps a very simple kind of "map", although the map is built from smells, angles, and edges, not lines on paper.

⑦ These three tools — pheromones, the sun, and landscape memory — can work together, or one can take over when another fails. On a cloudy day, a forest ant may rely more on the trail and on the shape of trees. On a shining desert at noon, the sun and step-counting may matter most. This mix of strategies is one reason ants are so successful: they do not depend on a single system.

⑧ There are other surprising details. Some ants have a tiny patch of **magnetic material** in their bodies, which helps them sense the Earth's magnetic field — almost like a small built-in compass. Other ants, when they come back to the nest, "teach" their sisters the route by walking slowly in front of them, stopping often, so their friends can catch up. A few tropical species even **count the dome of the sky**, using the pattern of light to find direction, the way sailors once used the stars.

⑨ Why should we care about all this? First, it is a reminder that intelligence is not only a matter of big brains. A colony of ants, each with a simple head, acts together as a kind of clever "super-organism" that can solve real problems — finding food, rebuilding a home, even escaping floods. Second, studies of ants have already helped computer scientists design better systems for delivery trucks, internet routers, and disaster-rescue robots. The rules ants use are simple, local, and flexible — exactly the kind of rules big machines often need.

⑩ So next time you see that neat line of ants on your kitchen floor, pause for a moment before you reach for a cloth. You are watching a quiet miracle of chemistry, geometry, and group memory. No one ant knows the whole plan. And yet, between them, the road back home is always clear.

根据文章内容，选择最佳答案。

1. Which of the following is NOT listed in the passage as one of the three main tools ants use to navigate?

A. Chemical signals called pheromones.
B. The position of the sun in the sky.
C. Memory of small features of the landscape.
D. Reading traffic signs painted by humans.

2. How do pheromone trails form a "strong" ant highway?

A. One big ant makes the whole trail alone.
B. Each ant that follows the scent also leaves more pheromone, making the thin line gradually into a strong, confident line.
C. An ant queen orders every other ant to walk the same way.
D. The food itself gives off the scent.

3. What happens in the experiment where scientists move an ant to a new spot?

A. The ant flies back home with no problem.
B. The ant still walks in the same direction and for the same distance it planned before, so it arrives at a wrong place and looks confused.
C. The ant gives up and lies down.
D. The ant immediately finds the new nest.

4. Why are ants sometimes called a kind of "super-organism"?

A. Because each ant has a big brain and many hands.
B. Because a whole colony of ants, each with a simple brain, works together to solve real problems like finding food or rebuilding a home.
C. Because ants are much bigger than other insects.
D. Because ants never move from a single spot.

5. What is the writer's main attitude at the end of the passage?

A. Ants are dirty and must be killed quickly.
B. The behaviour of ants is a small daily miracle of chemistry, geometry and group memory that deserves quiet respect.
C. Ants are less clever than any other insect.
D. Human science will never learn anything from ants.

答案与解析

1.
【答案】D
【题型】细节理解题（选非）
【定位】第②段 & 全文
【解析】三种工具：信息素、太阳、地貌记忆；不含"人类交通标志"。故选 D。

2.
【答案】B
【题型】细节理解题
【定位】第③段
【解析】每只沿迹返程的蚂蚁都再留信息素 → 强化。A/C/D 均错。故选 B。

3.
【答案】B
【题型】细节理解题
【定位】第⑤段末
【解析】按原方向、原距离前行后困惑。A/C/D 均错。故选 B。

4.
【答案】B
【题型】细节理解题
【定位】第⑨段
【解析】"超个体"=群体合作解决问题。A/C/D 均错。故选 B。

5.
【答案】B
【题型】主旨大意题
【定位】第⑩段
【解析】把蚁行视为值得敬意的小奇迹。A/C/D 均偏。故选 B。

词汇注释（超纲词）

- **pheromone** [n.] 信息素
- **antenna / antennae** [n.] (昆虫的)触角
- **polarised light** 偏振光
- **path integration** 航迹积分（一种寻路方式）
- **super-organism** [n.] 超个体